

AI & MACHINE LEARNING

Visual Framework Portfolio Artifact

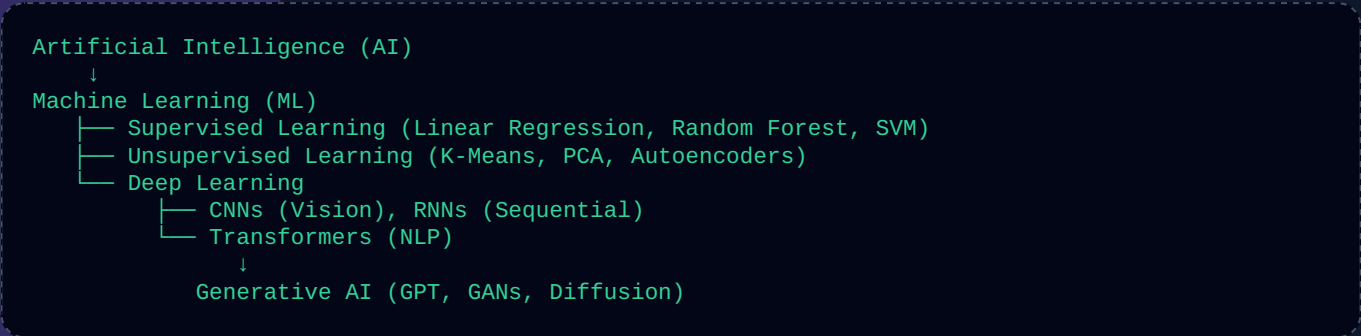
Artificial Intelligence is transforming industries through data-driven decision-making, automation, and content generation. ML, DL, and GenAI are connected technologies enabling systems to learn patterns and make intelligent predictions or create new content.

- **AI:** Broad field of intelligent systems
- **Machine Learning:** Subset of AI that learns from data
- **Deep Learning:** Subset of ML using neural networks
- **Generative AI:** Advanced DL models creating new text, images, and code

Major Machine Learning Algorithms

Algorithm	Type	Application Domain	Real-World Use Case
Linear Regression	SUPERVISED	Tabular Data	House prices, sales forecasting
Random Forest	SUPERVISED	Tabular Data	Fraud detection, customer churn
SVM	SUPERVISED	Vision/Tabular	Face recognition, spam filtering
K-Means Clustering	UNSUPERVISED	Tabular Data	Customer segmentation
PCA	UNSUPERVISED	Tabular Data	Data compression
CNNs	DEEP LEARNING	Computer Vision	Self-driving cars, medical imaging
Transformers	DEEP LEARNING	NLP	ChatGPT, sentiment analysis
GANs	GEN AI	Media Gen	Deepfakes, art generation
Diffusion Models	GEN AI	Computer Vision	Stable Diffusion image creation
Autoencoders	DEEP LEARNING	Vision/NLP	Anomaly detection, denoising

Hierarchy Taxonomy



Key Domains and Industry Use Cases

1. Tabular Data (Structured Datasets)

Algorithms: Random Forest, Linear Regression, K-Means.

Industry: Banking fraud detection, healthcare risk prediction, business analytics.

2. Computer Vision (Image & Video)

Algorithms: CNNs, GANs, Diffusion Models.

Industry: Autonomous vehicles, facial recognition, medical imaging diagnostics.

3. Natural Language Processing (NLP)

Algorithms: Transformers (GPT, BERT), RNNs.

Industry: Virtual assistants (ChatGPT), machine translation, sentiment analysis.

4. Generative AI (Content Creation)

Algorithms: GPT Models, GANs, Stable Diffusion.

Industry: AI-generated art, automated content creation, music generation.

Reflection

Creating this framework allowed me to connect specific algorithms with real-world applications. Supervised learning excels with labeled data, while deep learning dominates vision and language tasks. I realized that while Generative AI is growing rapidly, traditional models like Random Forests remain vital for business analytics. This project strengthened my appreciation for how AI technologies continue to evolve across healthcare, finance, and education.

References

- IBM - What is AI / Generative AI / Capability Model
- IBM Blog - AI vs ML vs DL vs Neural Networks
- IBM Technology - AI, ML, DL and GenAI Explained (YouTube)
- Industry Discussion - Traditional ML vs GenAI (Reddit)